# Java Stream API - Problem Solving Interview Exam

**Questions:** 10 **Difficulty:** Intermediate to Advanced **Duration:** 90 Minutes

**Instructions:** Solve all questions using Java 8+ Stream API. Avoid traditional loops.

## Question 1: Find the First Non-Repeated Character

Given a string, find the first character that appears only once using the Stream API.

**Input:**

```
String input = "swiss";
```

**Expected Output:**

```
w
```

## Question 2: Find Duplicate Elements in a List

Given a list of integers, find all duplicate elements using the Stream API.

**Input:**

```
List<Integer> numbers =
    Arrays.asList(10, 20, 30, 20, 40, 10, 50, 30);
```

**Expected Output:**

```
[10, 20, 30]
```

## Question 3: Find the Second Highest Number

Given a list of integers, find the second-highest distinct number using the Stream API.

**Input:**

```
List<Integer> numbers =
    Arrays.asList(10, 5, 20, 20, 8, 15, 10);
```

**Expected Output:**

```
15
```

## Question 4: Find the Frequency of Each Word

Given a sentence, count the frequency of each word using the Stream API. The comparison should be case-insensitive.

**Input:**

```
String sentence = "Java Spring Java Hibernate Spring Java";
```

**Expected Output:**

```
java=3
spring=2
hibernate=1
```

## Question 5: Find the Longest String

Given a list of strings, find the longest string using the Stream API. If multiple strings have the same maximum length, return the first one.

**Input:**

```
List<String> words = Arrays.asList(
    "Java", "SpringBoot", "Microservices",
    "Hibernate", "Docker"
);
```

**Expected Output:**

```
Microservices
```

## Question 6: Group Employees by Department

Given a list of employees, group them by department and calculate the average salary for each department using the Stream API.

**Input:**

```
class Employee {
    String name;
    String department;
    double salary;
}

Ahmed   IT       10000
Mohamed IT       15000
Sara    HR        8000
Omar    HR       12000
Ali     Finance  20000
```

**Expected Output:**

```
IT      = 12500.0
HR      = 10000.0
Finance = 20000.0
```

## Question 7: Find the Highest-Paid Employee in Each Department

Given a list of employees, find the employee with the highest salary in each department using the Stream API.

**Input:**

```
class Employee {
    String name;
    String department;
    double salary;
}

Ahmed   IT       10000
Mohamed IT       18000
Sara    HR        9000
Omar    HR       14000
Ali     Finance  20000
```

**Expected Output:**

```
IT      = Mohamed
HR      = Omar
Finance = Ali
```

## Question 8: Find Common Elements Between Two Lists

Given two lists of integers, find their common distinct elements using the Stream API. Preserve the order in which they appear in the first list.

**Input:**

```
List<Integer> list1 =
    Arrays.asList(1, 2, 3, 4, 5, 6, 3);
List<Integer> list2 =
    Arrays.asList(4, 5, 6, 7, 8, 4);
```

**Expected Output:**

```
[4, 5, 6]
```

## Question 9: Find the Top 3 Most Frequent Numbers

Given a list of integers, find the three most frequently occurring distinct numbers using the Stream API. Sort by frequency descending; break ties by numerical value ascending.

**Input:**

```
List<Integer> numbers = Arrays.asList(
    1, 2, 2, 3, 3, 3, 4, 4, 4, 4,
    5, 5, 5, 6
);
```

**Expected Output:**

```
4 = 4 occurrences
3 = 3 occurrences
5 = 3 occurrences
```

## Question 10: Find the Top 3 Highest-Paid Employees in Each Department

Given a list of employees, find the top three highest-paid employees in each department using the Stream API. Sort by salary descending; break ties alphabetically by name.

**Input:**

```
class Employee {
    String name;
    String department;
    double salary;
}

Ahmed    IT       10000
Mohamed  IT       18000
Ali      IT       15000
Omar     IT       22000
Sara     HR       12000
Mona     HR       16000
Nour     HR       14000
Hana     HR        9000
Khaled   Finance  20000
Youssef  Finance  25000
Mostafa  Finance  18000
```

**Expected Output:**

```
IT:
Omar     22000
Mohamed  18000
Ali      15000

HR:
Mona     16000
Nour     14000
Sara     12000

Finance:
Youssef  25000
Khaled   20000
Mostafa  18000
```